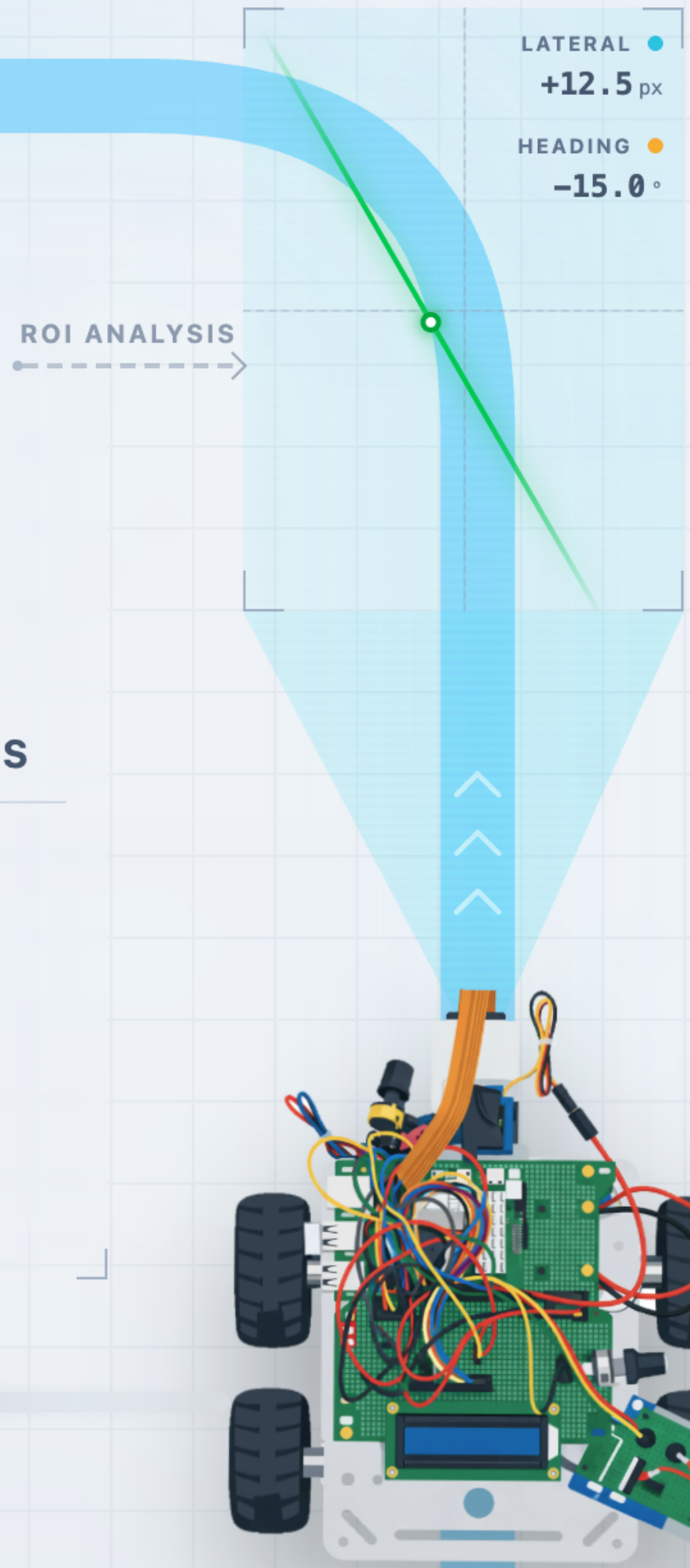
GROUP5.

APPLIED ENGINEERING PROJECT

PROTO-04



VISION SYSTEM

- Real-time lane detection via **cv::fitLine**
- Line loss memory
- Lateral offset & heading angle computation

SYSTEM SPECIFICATIONS

SOFTWARE	C++ / OpenCV
HOST	Raspberry Pi
VISION	Camera Module 3
MOTION	1-DOF SG90
COMMS	UART Driver Board
CHASSIS	4WD Differential
POWER	18650 × 2

GROUP MEMBERS

Luxuan SUN
Junyang ZHANG
Zichao MENG
Zihao ZHAO

NIGHWAN

AUTONOMOUS VISION-BASED GROUND VEHICLE